airtel

SYNC

ENTERPRISE TRACK A · PROJECT UDYAM DIGITAL

Udyam Digital

A small business goes live in one sitting: on a phone, in Hindi or English, with no salesperson in the loop.



Team IIITD

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LIVE PROTOTYPE udyam-digital-isyfatadrab4adtbqddzwp.streamlit.app

PROTOTYPE VIDEO Watch the 3-minute demo on Google Drive

CODE ON GITHUB github.com/Sheel23501/udyam-digital

₹35.6 lakh of ₹55.4 lakh went to leads that never activated

4,632 of 5,000 leads dropped last quarter. Half of those drops were the sales process itself, not the offer.

48%

of drops: unreachable, no time for a visit, visit not scheduled, quote too slow or too complex

₹53,026

field-sales cost per activation on micro retail, for a ₹2,619/month customer

17 days

median lead-to-activation; the quote alone takes 4 days

719 → 9

owners asked for 5G AirFiber 719 times; sales sold it 9 times

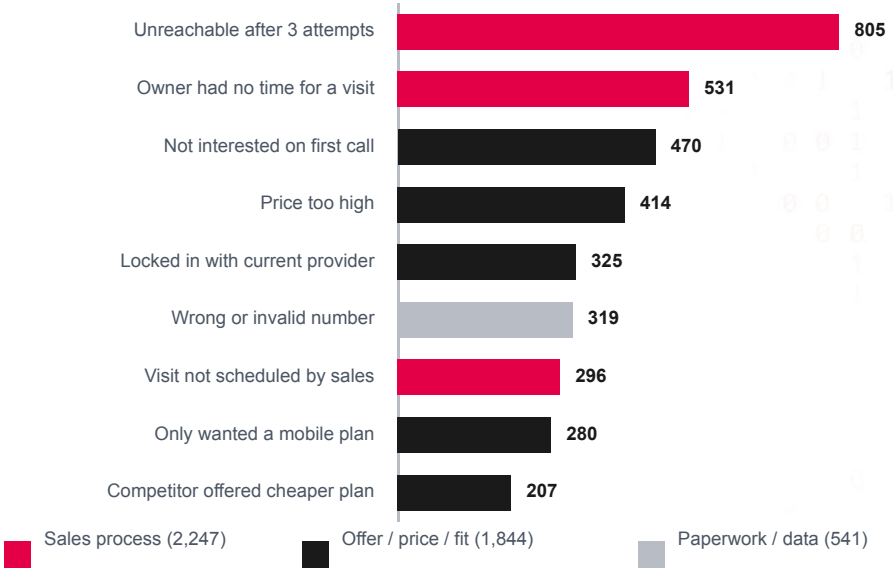
22

customers activated last quarter with a cancelled or suspended GSTIN

6% → 28%

6-month churn as the status calls a customer had to make rose from 1 to 5+

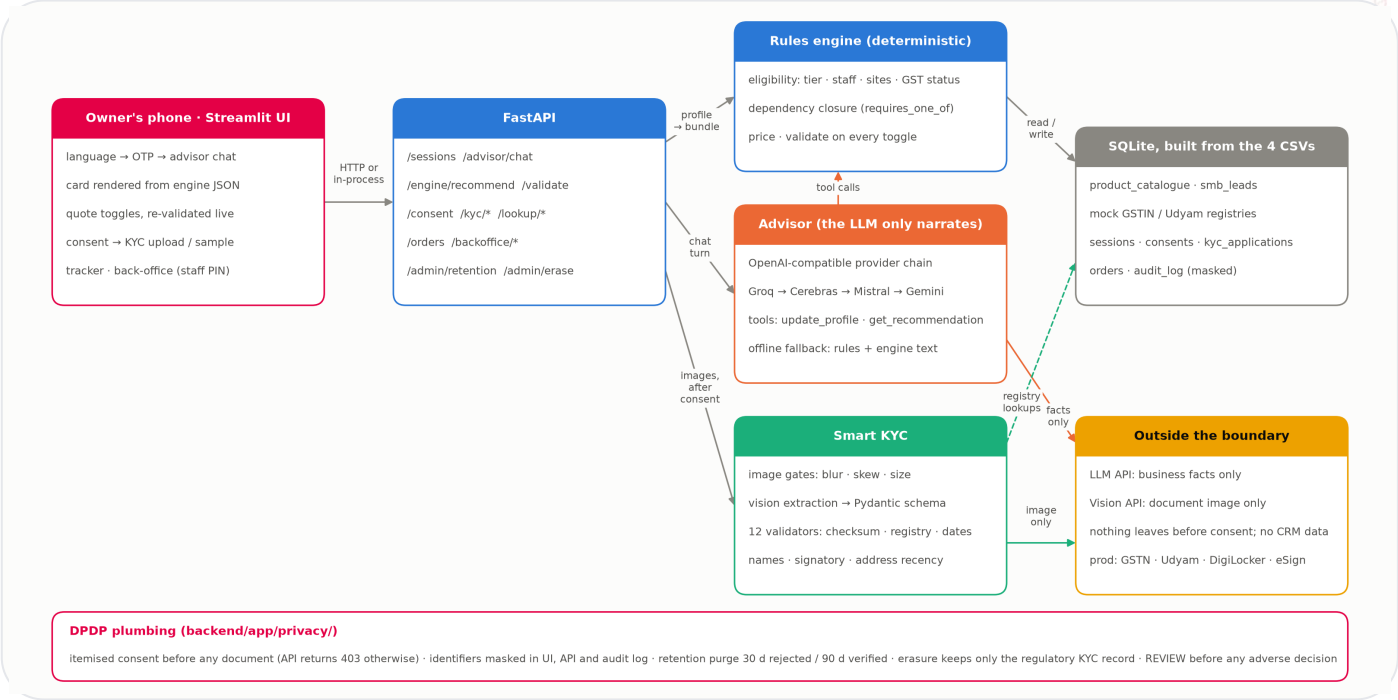
Why 4,632 leads dropped (top 9 reasons)



Source: smb_leads.csv, 5,000 leads (2026-04-01 to 2026-06-30); every figure reproduced by backend/scripts/make_charts.py.

Our approach: rules decide, the LLM narrates

1. **Enter**
   link, QR or WhatsApp; mobile + one-time code
2. **Advisor**
   4 to 6 questions in Hindi or English
3. **Quote**
   one monthly number, re-checked on every toggle
4. **Consent**
   itemised DPDP notice before any document
5. **Smart KYC**
   four photos, 12 checks in seconds
6. **Tracker**
   dated stages, zero status calls



Consent gates every document; the engine still quotes if the model is down. Full-resolution diagram: docs/architecture.png (submitted separately).

Smart KYC: 12 checks in seconds, built around the DPDP Act

Vision extraction of four documents, then deterministic validators against the GST and Udyam registries. A photo problem is a REVIEW, never a rejection.

The 12 sample applicants as the prototype grades them: 12/12 match the key

FAIL **K01** GSTIN cancelled; Udyam number missing	REVIEW **K02** GST certificate photo rotated	PASS **K03** all documents consistent
REVIEW **K04** GST certificate photo blurred	FAIL **K05** Udyam certificate missing	FAIL **K06** GSTIN check digit fails: tampered digit
FAIL **K07** legal name differs across documents	PASS **K08** all documents consistent	PASS **K09** all documents consistent
FAIL **K10** signatory is not the person of record	FAIL **K11** electricity bill 257 days old (limit 90)	PASS **K12** all documents consistent

22 → 0

activations with a cancelled or suspended GSTIN: the registry is checked live, before the order

DPDP controls, each a code path, not a slide

Consent gate: documents are refused (HTTP 403) until the owner ticks four itemised purposes in their language
Minimisation: five facts for the quote; GST or Udyam path; no Aadhaar; only the image leaves the boundary
Masking: GSTIN as 27XXXXXXXXXX1ZP, names as 'Nisha A.' in UI, API and audit log
Retention: images deleted 30 days after a rejection, 90 after verification; erasure on request
Accuracy: REVIEW before any adverse decision; 'talk to a human' always available

Reproduce: cd backend && python scripts/kyc_report.py --key tests/kyc_answer_key_images.json. Privacy note: docs/privacy-note.md (submitted separately).

Competitive analysis: where Udyam Digital differs

Public sources only, reviewed November 2026. "Not public" means we found no published detail, not that the feature is absent.

	Airtel today	Jio Business	Vi Business	Tata Tele (TTBS)	Udyam Digital
How an owner starts	Field visit or call-back. 48.5% of drops are the sales process	Plan page, then business details and KYC documents online (third-party summary)	Online form, then a rep calls back within 48 h (corporate postpaid)	Sales-led. No public self-serve flow found	**Self-serve on the owner's phone, in Hindi or English**
Price	Quote after a visit. The quote alone takes 4 days	Tiered plans, reported from ₹501 at the 2021 launch	Postpaid from ₹299 a line	Pay-as-you-grow. Prices not public	**One monthly number in the chat, re-checked on every toggle**
What is bundled	25 products, sold one by one	Connectivity plus apps: Jio Online, Teams and Meet licences (2021)	Mobile plans, data pooling for small firms	Connectivity, cloud calling and security suites	**A rules-checked bundle per segment with dependencies closed**
KYC	Paper at the visit. 22 activations had a non-active GSTIN	GSTIN or PAN documents (third-party summary)	Assisted KYC at the doorstep. A DoT notice reported on business lines without end-user KYC	*Not public*	**12 checks in seconds against GST and Udyam, after consent**
Language and tracking	Status by phone: 3.0 calls per customer	*Not public*	*Not public*	*Not public*	**Hindi and English. Dated tracker, zero status calls**

None of the public material we reviewed shows one phone journey from quote to registry-checked KYC to a validated bundle.

Sources: businesstoday.in (Jio, Mar 2021), myvi.in corporate postpaid pages, bharatnet.in (DoT notice), tatatelebusiness.com and tele.net.in. Airtel and Udyam Digital columns: smb_leads.csv and the prototype.

Impact: 12% activation at half the cost per customer

Two-quarter targets on the same lead volume and acquisition spend, measured weekly from tables the prototype already writes.

Lead to activation

7.4% → 12%

Cost per activation

₹15,053 → ≤₹7,500

Lead to live

17 days → ≤5 days

Status calls per customer

3.0 → under 0.5

6-month churn

15.2% → ≤10%

Non-active GSTIN activations

22 → 0

Who we win first, and what they buy

Segment	Bundle, from ₹/month	Payback
Small multi-site	Multi-Store ₹21,579	0.5 mo
Small manufacturers	Factory Connect ₹14,377	0.8 mo
Professional offices	Clinic / Office Pro ₹5,487	1.6 mo
Micro retail & services	Dukaan Start ₹1,198	8.0 → 1.2 mo

Micro retail goes digital-only: at a ₹3,000 cost per activation its payback falls from 8.0 to about 1.2 months.

What it is worth. The same ₹55.4 lakh at 12% activation brings 600 customers a quarter instead of 368, 232 more. At today's segment mix and ₹12,351 average monthly value, that is about ₹28.7 lakh more monthly recurring revenue, before churn.

Projection from the Apr to Jun 2026 leads file; it assumes the extra customers match today's segment mix. Targets are ours, not Airtel commitments.

Rollout, risks and proof

Start where the data says the money is, and say plainly what could go wrong.

Q3 2026 **Prove it**	Delhi and Mumbai, Hindi and English. Micro retail and professional offices. Live GSTN and Udyam APIs. Partner-code pilot. **Gate: 12% activation, quote in under 5 minutes**
Q4 2026 **Widen**	Tier-2 cities. Multi-site journeys. DigiLocker and eSign. Field team routed by the propensity queue. **Gate: cost per activation ≤ ₹7,500, KYC first pass ≥ 70%**
Q1 2027 **Deepen**	Manufacturers with field assist for leased line and SD-WAN. Serviceability at quote time. Regional languages. **Gate: 6-month churn ≤ 10%, status calls under 0.5**

Risk	What we built against it
The model suggests something the owner cannot have	The rules engine decides every item. 30 scenarios and 16 live chats, zero violations
KYC wrongly rejects a genuine owner	Photo problems become REVIEW with a plain fix. Every FAIL has a reason and a human hand-off
Telecom end-user KYC still applies to SIMs	Registry checks are added to the DoT-required step, not a replacement. A reported DoT notice to Vi shows the exposure
Model or API down mid-journey	Provider chain, cached replies, offline engine. The journey completes with no model
Channel conflict with partners and field sales	Partner code on every journey. High-value leads routed to humans
DPDP exposure	Consent gate, masking, retention purge, erasure, audit log

2,050 automated tests green in CI

30 / 30 advisor scenarios pass

12 / 12 KYC verdicts match the key

16 / 16 live chats, zero violations

Live prototype and code are linked on the title slide. Step-by-step checks: docs/how-to-check.md. Privacy note and architecture diagram are submitted separately.